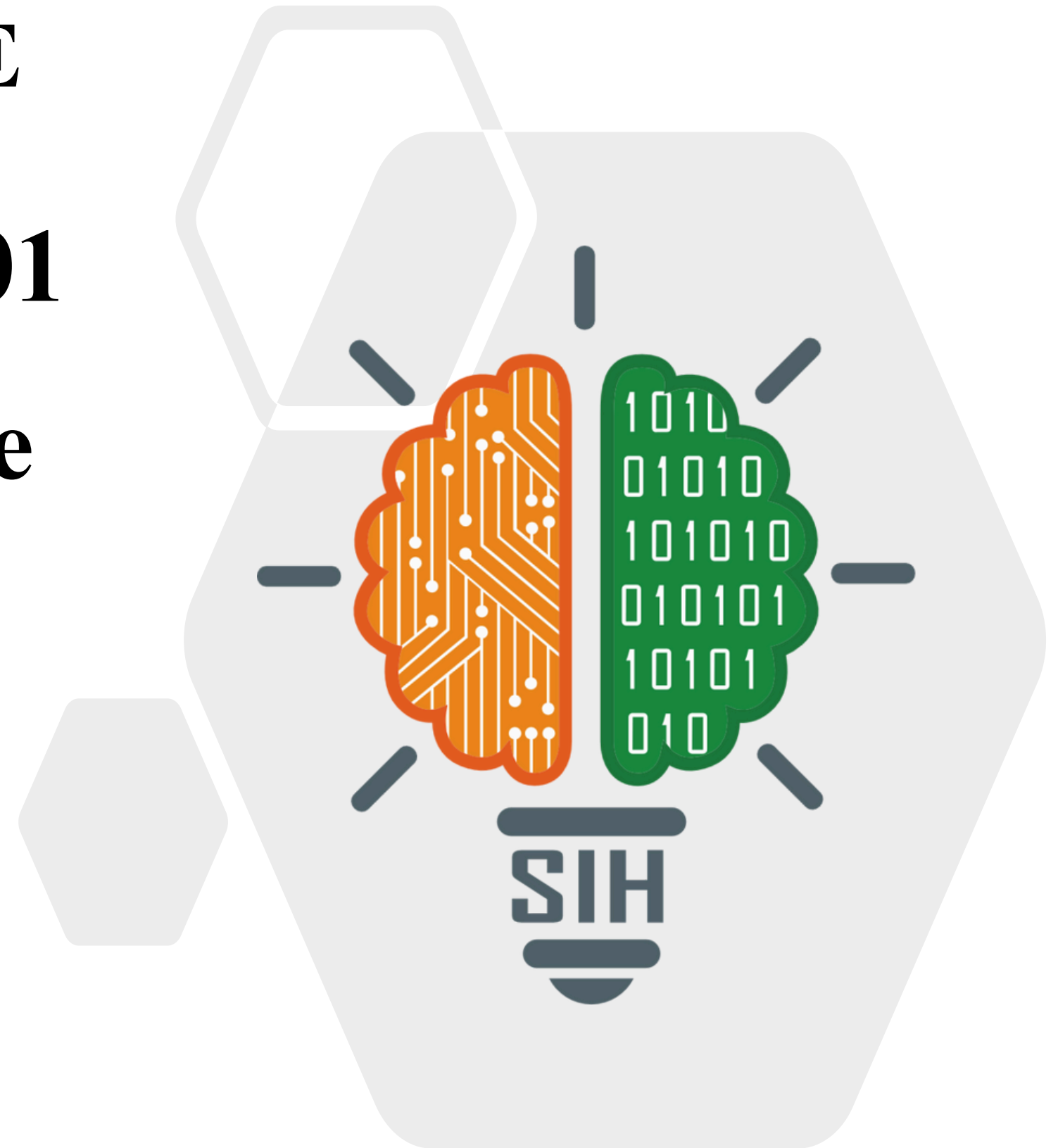


TITLE PAGE

- **Problem Statement ID- SIH25101**
- **Problem Statement Title- Remote classroom for rural colleges**
- **Theme- Smart Education**
- **PS Category- Software**
- **Team Name- Code Diggers**



Rural colleges often face faculty shortages and poor connectivity, leaving students at a disadvantage. Our solution is a lightweight virtual classroom app that combines live and recorded teaching, functions reliably in low bandwidth, and supports both online and offline learning. It bridges the gap by connecting city-based experts with rural campuses, ensuring uninterrupted education without costly infrastructure. What sets it apart is its context-aware design optimized for low data usage, simple devices, and multilingual access making education inclusive and sustainable for rural colleges.

STUDENT FEATURES :

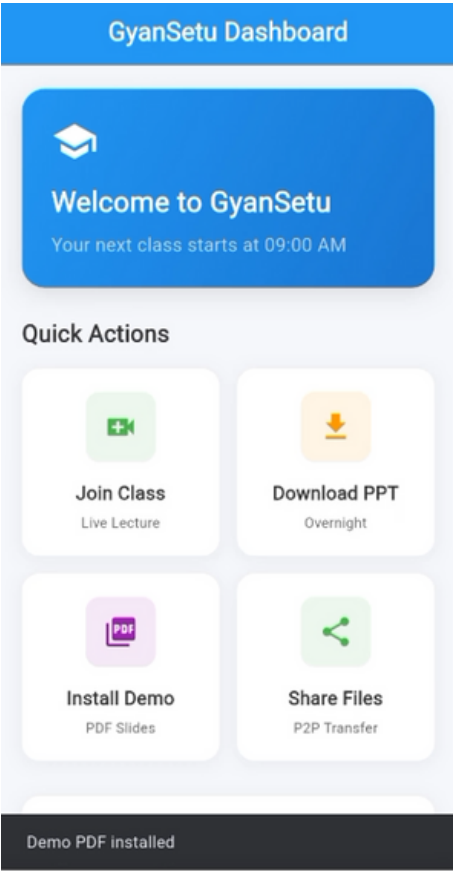
- **Smart Caching**
compressed slides and audios and caching continues even if internet drops
- **Audio Quality**
our app focuses on clear audio. During live classes, sound is compressed smartly so it stays smooth and clear.
- **Quizzes**
JSON in app quizzes for engagement
- **Overnight Downloads:**
lectures and microcapsules automatically download during off-peak hours, minimizing data costs.
- **P2P Sharing:**
offline sharing via Bluetooth/nearby helps peers without connectivity access lectures.

TEACHER FEATURES :

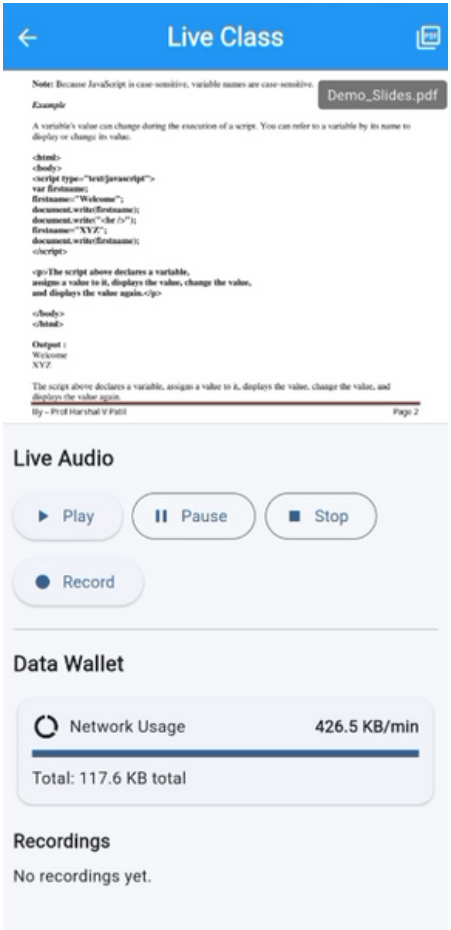
- **Teaching Resources**
teachers can upload slides and record audios seamlessly
- **Class Management**
schedule and conduct sessions with ease
- **Analytics & Tracking**
monitor attendance, quiz results and engagement

COMMON FEATURES :

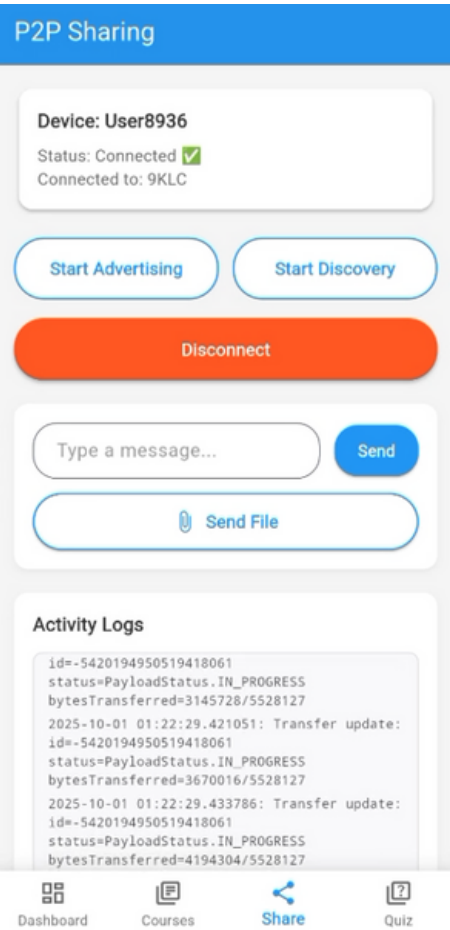
- **Multilingual UI**
simple and role-adaptive interface
- **Smart Data Wallet**
precise tracking of bandwidth usage for lectures and quizzes



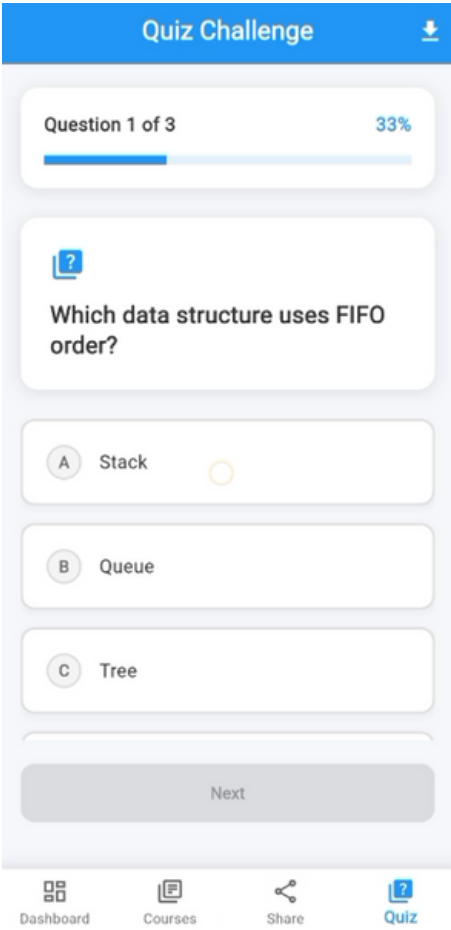
Dashboard - Quick access to join classes, download PPTs and share files.



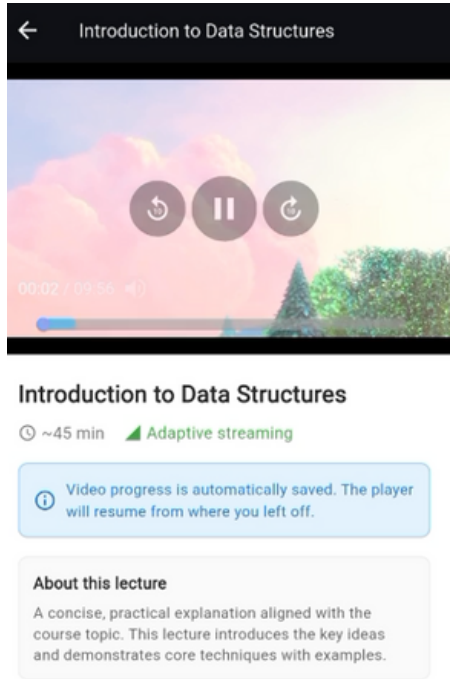
Live Class - Low-bandwidth audio-first streaming with synced slides and data wallet tracking.



P2P Sharing - Offline lecture and file transfer between students without internet.

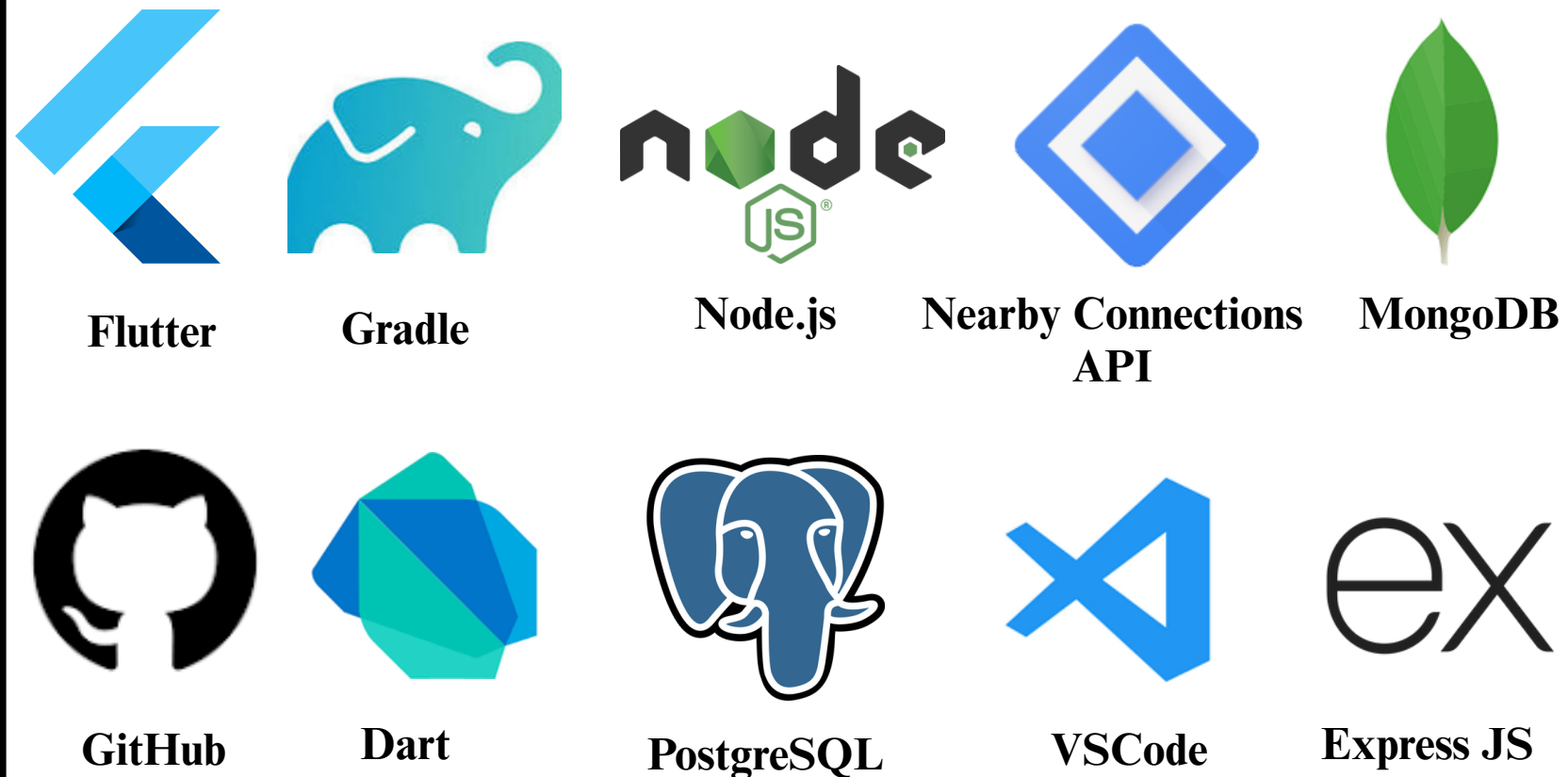


Quiz Challenge - Interactive quizzes in JSON format for engagement.



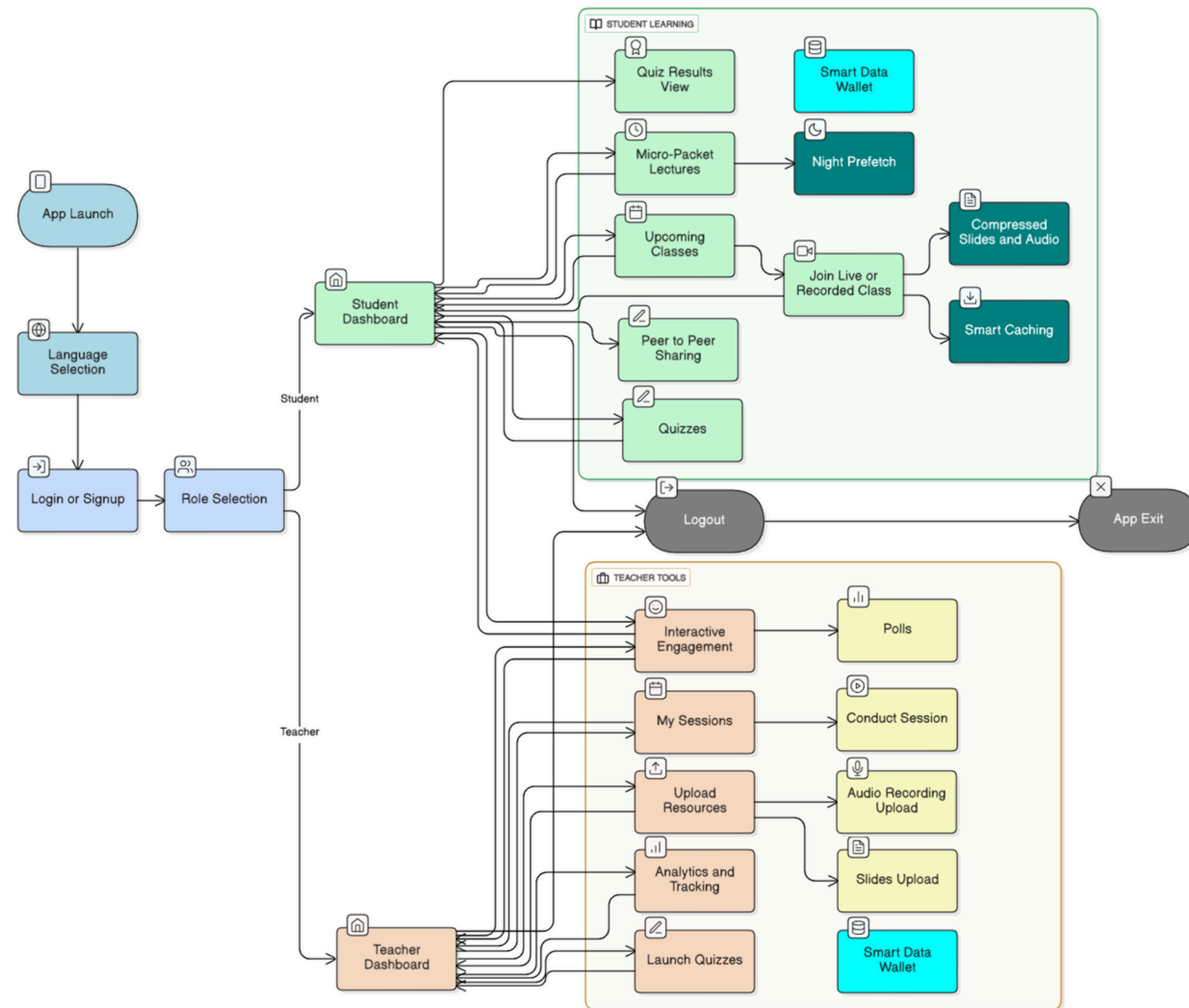
Micro-Lecture Capsule - Recorded session split into 15-min packets with offline access.

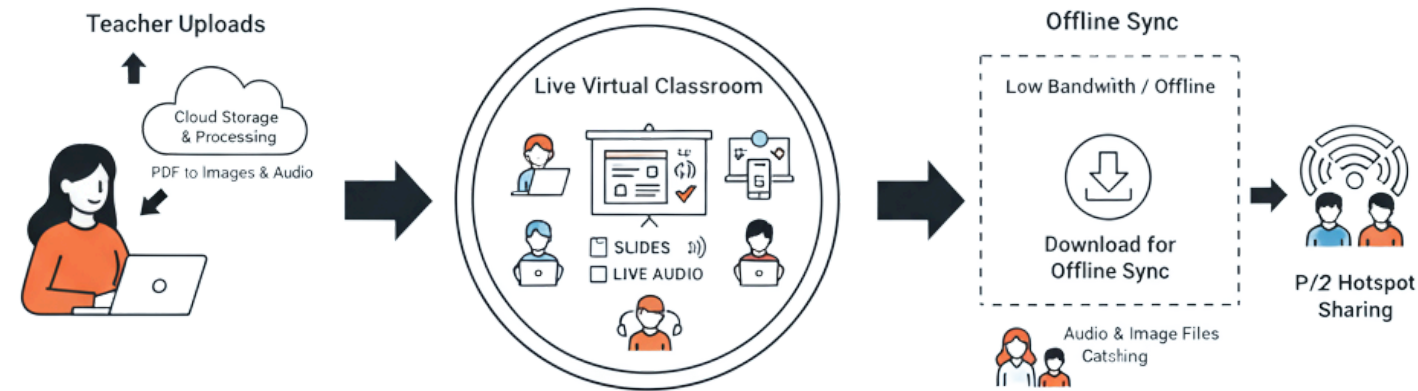
Technology Stack:



Product Status:

70% of the product has been built and development is ongoing. Testing and validation has been done.



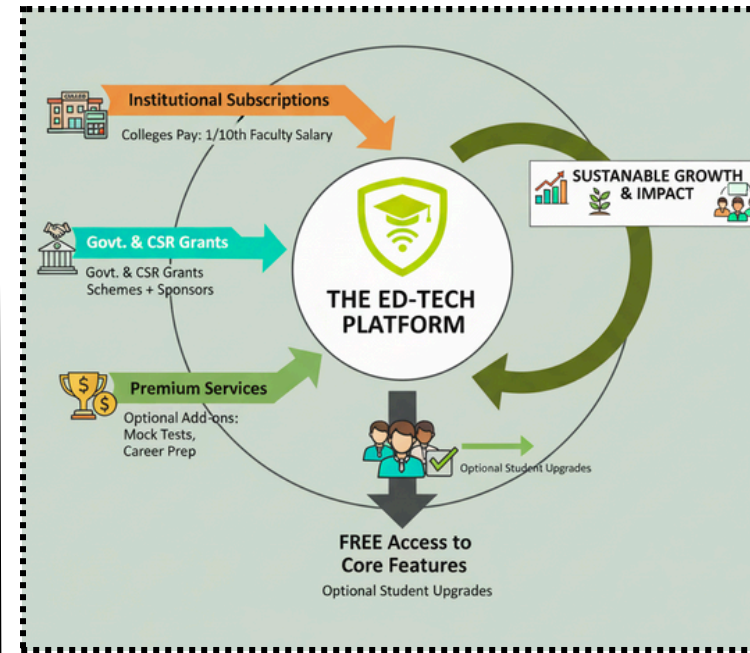


- Low Bandwidth → video buffering / class failure**
 - sol : compressed audio +PPT sync, night auto-download for offline access
- Limited Devices → students may not own laptops**
 - sol : Mobile-first design; works on low-cost Android entry level phones.
- Faculty may not be tech-savvy**
 - sol : simple recording/upload interface, requires minimal learning curve.
- Limited IT staff in rural colleges.**
 - sol : centralized backend hosted in cloud, small caching server needed locally

Feature / Aspect	Our App	Zoom
Low Bandwidth Support	Lossless audio + slides; works on 2G/3G	Requires stable high-speed internet
Offline Access	Downloadable sync file + cached content	Not available
Device Requirements	Mobile-first; works on low-cost smartphones, low storage req.	Mostly laptops/smartphones, high storage req.
P2P Hotspot Sharing	Supported to reduce data cost and improve access	Not supported
Technical Infrastructure	Minimal local setup; centralized cloud backend	Requires stable network & institutional IT support

Revenue Model

- Institutional Subscriptions** : Annual fees paid by colleges, 1/10th cost of full time faculty → affordable and stable revenue
- Government & CSR funding** : Grants and social impact funds for full-scale deployment
- Optional Premium Add-ons (B2C)** : Freemium advanced features (career guidance, mock tests) for long-term growth, keeping core education free for students.



Scalability

Cloud-Native Infrastructure

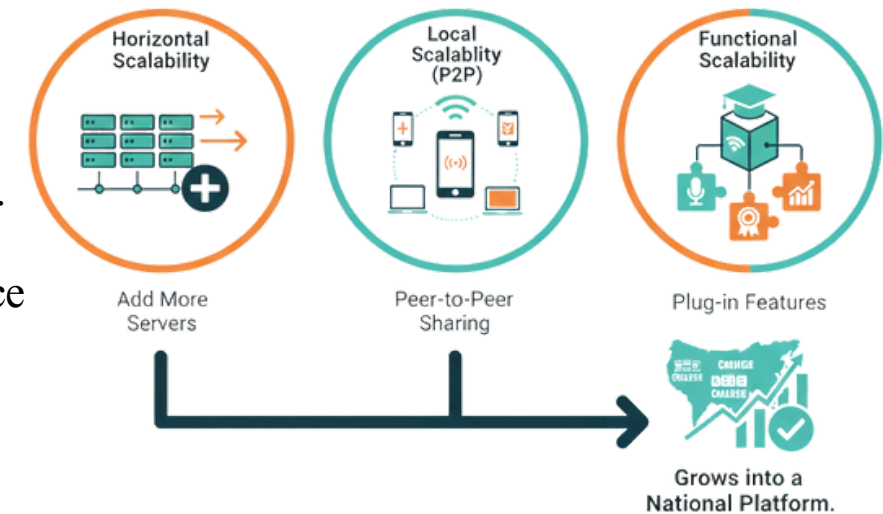
- Resources (servers, DB, bandwidth) scale up/down automatically with demand.
- No dependency on costly local infrastructure.

Efficient Content Delivery

- Compression, caching, and offline sync reduce data load.
- Keeps performance stable even in low-bandwidth regions.

P2P & Hotspot Sharing

- Students share content locally, easing pressure on central servers.
- Growth in users doesn't mean linear growth in server load.



40 M+ STUDENTS

impacted in rural higher education

80% lower data use
<26 mb app size
works in 2G/3G networks

1 TEACHER = 1000 STUDENTS

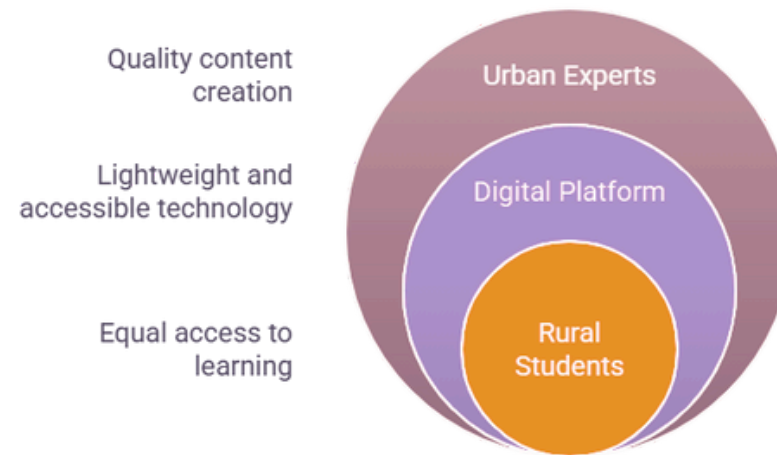
Bridges the faculty shortage gap (65% unfilled positions)
with remote teaching.

Cuts content delivery time by 70%

100 Cr+ POTENTIAL SAVINGS

annually in infrastructure/data.

7,000+ diploma colleges across India → scalable nationwide.
10% adoption = 4M students impacted in year one.



impact diagram



remote classrooms



Bridges urban-rural divide

- equal access to advanced courses like AI, VLSI
- removes dependency on travel and unreliable video platforms



Designed for entry level smartphones

- 26 mb app, 10x lighter than a single zoom recording
- optimized for budget phones in rural india.
- compressed visuals + cached recordings for low storage



Affordable and Data Efficient

- 80% less data usage than youtube lectures
- Off-peak downloads + JSON quizzes → minimal cost
- sustainable for families with prepaid data models



Reliable learning, even Offline

- cached recorded lectures available offline
- peer-to-peer (P2P) file sharing saves data between students in the same campus
- works in villages with patchy 2G/3G networks



Empowered Faculty

- no costly equipment for giving lectures.
- simple UI/UX reduces learning curve for teachers
- enables experts from cities to teach rural classrooms easily



Scalable National Impact

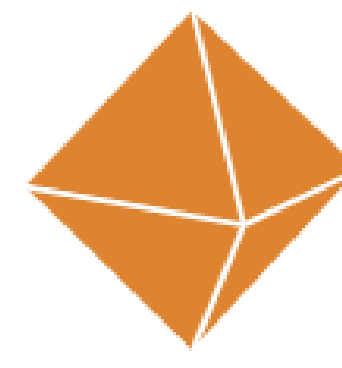
- Blueprint for inclusive education at scale.
- Potential to reach 7,000+ diploma colleges & millions of students.
- Permanent transformation of India's rural higher education ecosystem.

- Approximately **42%** of rural students lack **reliable internet access**, hindering online education. - [mediumlink](#)
- A study by the Azim Premji Foundation in 2021 indicated that nearly **60%** of school children in India **cannot access online learning** opportunities. - [avenues&challenges](#)
- Students in rural areas often face issues such as limited access to digital devices and **inconsistent internet connectivity**, impacting their ability to participate in online education. - [irjmets](#)
- UDISE+ 2023–24 (Unified District Information System for Education Plus) – shows ~68.3% schools in Rajasthan had internet, 51.6% had computers but **30-35% still lacked connectivity**.
- Annual Status of Education Report (ASER) 2021 – found that **1.43 crore** children in Bihar lacked digital access for online learning during COVID.
- Study by Bihar Economic Survey 2021-22 / local news summaries – In Patna & Muzaffarpur, nearly **90% students** did not access online education during lockdowns
- UDISE+ 2023–24 – Reports **only 38.6%** of schools in UP had internet facilities.



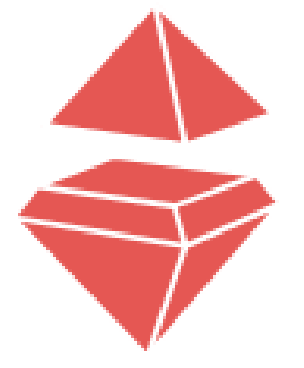
Global Evidence

Worldwide remote learning gaps



India Evidence

ASER, SCHOOL survey, DU/State stats



Our App Model

Low-bandwidth, offline-first, P2P